



META CIRCLE RESEARCH

Beyond Relative Generalization Invariance: Shared Difficulty and Local Learning Response in Language Model Training

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Relative Generalization Invariance describes approximately preserved token-wise loss differences across some language-model training settings. We locate the terms behind loss similarity using a common late checkpoint and teacher-forced continuations. The measured decomposition is $L = D + A + Q_2 + \varepsilon$: frozen batch difficulty, frozen-gradient work on actual parameter motion, full-vocabulary logit variance, and a checked remainder. Six settings, including normalized momentum, Adam, and output-head-only training, restart from Pythia-70M step 143000 with a shared stream and 32-step warmup. Three paired permutations are evaluated under three explicitly adaptive rate protocols, retaining 54 completed trajectories. At the terminal scale, shared difficulty explains over 99.9926% of raw arriving-loss variation. Adam and the affine head-only control pass the 10% response gate, while full-model SGD/momentum still fail at 10.51–15.21%, predominantly through network nonlinearity. The head-only intervention removes that component to numerical precision, and larger-rate settings expose softmax approximation failure. An exact same-stream momentum identity connects optimizer memory to low-step alignment. Nevertheless, endpoint displacement cosine above 0.9997 does not produce a constant nonzero token-loss gap. The results identify measured local contributions and substantive boundaries of a universal RGI explanation, rather than infer one from raw-loss correlation.

Keywords: language model training, teacher forcing, optimization dynamics, relative generalization invariance, cross-entropy

Project page: <https://guoshaoyang-pku.github.io/RGI-explained/>

1. Introduction

Language-model loss reflects both the batch that arrives and the motion of the predictor. The same difficult targets can raise several models’ losses together, while the training method changes how their predictions move. A useful explanation should locate both contributions on the same teacher-forced targets and verify which dominates at the scale being measured.

Relative Generalization Invariance (RGI) (Zhang et al., 2026) asks why loss differences across some training settings remain approximately preserved across validation tokens. A complementary study (Lee and Lee, 2026) finds near-overlapping low-rate loss and sharpness traces for normalized linear gradient filters. Neither scalar trace agreement nor a stylized construction identifies which sample-conditioned terms dominate a real Transformer continuation. High raw-loss correlation also need not imply a constant token-wise gap.

We make the mechanism measurable. All branches start from the same late Pythia-70M checkpoint, reset optimizer state, re-warm up, and see the same prescribed batch sequence. The frozen checkpoint gives difficulty D ; the actual parameter displacement gives frozen-gradient work A ; and the full-vocabulary logit displacement gives nonnegative variance Q_2 . We measure the remainder ε rather than assume it negligible. The central table and four-panel curves show these terms separately, without fitted slopes, intercepts, or time warps.

Three questions are distinct: whether D dominates raw arriving-loss variation, whether $A + Q_2$

closes $L - D$, and whether a paired gap is constant across tokens after D cancels. We test each with its own prespecified denominator and gate, retaining failed settings and evaluating a fixed held-out probe alongside the arriving stream. Our output-head-only control makes logits affine in the updated parameters and removes network nonlinearity exactly. A same-stream SGD/momentum identity separates gradient feedback from optimizer memory and warmup.

The claims are local. Rate extensions are disclosed as adaptive follow-ups to failed long-displacement approximations, not an untouched confirmation. These finite Pythia-70M continuations are not reproductions of the original 124M and 0.7B pretraining runs or a universal Transformer law.

2. Relation to RGI and scope of the claim

RGI compares a pair of models on the same teacher-forced target and asks whether their loss difference is approximately independent of the token. We retain that paired evaluation, but separate two notions that are easy to conflate. A weak statement is that two loss vectors have high correlation or a small relative-difference statistic. A strong statement is that

$$\ell_A(x, y) - \ell_B(x, y) \approx c \quad \text{for every evaluated prefix-target pair,} \quad (1)$$

with a residual small relative to $|c|$. The latter is the constant-offset mechanism that would make the optimizer choice nearly irrelevant to token ranking.

Our experiment is designed to locate terms before testing Eq. (1). On a natural target, an arbitrary checkpoint reference q is not the unknown conditional data distribution. Therefore a reference KL is a measurable comparison to q , not a natural Bayes excess risk. For a fixed teacher distribution we can remove this ambiguity, but that changes the task to a known-teacher continuation. We report the two settings separately.

The prior quadratic construction is a sufficient example in a different model class. It does not by itself identify the representation term or optimizer response of a Transformer. Our evidence likewise does not claim that natural LLMs satisfy the sufficient conditions of that construction. Instead, it tests whether the more concrete finite-displacement terms below account for the observed slow response and then checks whether the remaining paired gap is actually constant.

Relation to normalized gradient flow. Lee and Lee (Lee and Lee, 2026) report nearly coincident low-learning-rate loss and sharpness traces for fixed linear gradient filters on CIFAR-10, after normalizing the effective step $h = \eta Q(1)$ and progress $\tau = th$ by the filter’s dc gain. Their local modified-flow expansion, for a smooth objective and stable filter with slowly varying gradients through its memory, is

$$\frac{d\theta}{d\tau} = -g - h\alpha_Q Hg + O(h^2), \quad \alpha_Q = \frac{1}{2} - \frac{Q'(1)}{Q(1)}. \quad (2)$$

The leading flow is shared while the filter shape changes the next correction. Their consecutive-gradient cosine compares adjacent iterations inside a run, not parameters across optimizers. Preconditioning is outside that study’s scope; Adam is consequently an additional empirical intervention here, rather than an instance of its scalar dc normalization.

Shared leading flow does not imply a nonzero constant token-wise loss offset. Even nearby points on one flow have a first-order loss difference proportional to $g_i^\top g_{\text{train}}$, which depends on the scored token i . The proposed connection is therefore precise: low-step alignment may constrain the response differences in Eq. (11), while their centered sample structure still requires direct measurement. Our rate labels describe a controlled rate contrast; without measured stability geometry they do not identify an edge-of-stability regime.

3. A finite-update decomposition

Consider one prefix-target pair (x, y) , with logits $z_0(x)$ at the frozen start θ_0 , probabilities $p_0 = \text{softmax}(z_0)$, and logits z_t after the observed continuation. Write $\Delta z = z_t - z_0$, $p_t = \text{softmax}(z_t)$,

and e_y for the one-hot target. For a batch, all quantities below are averaged over the same target positions.

The frozen difficulty is

$$D(B) = \text{mean}_{(x,y) \in B} [-\log p_0(y | x)]. \quad (3)$$

The exact change in cross-entropy has a useful logit-space identity,

$$L_t(B) - D(B) = W(B) + Q(B), \quad (4)$$

$$W(B) = \text{mean} \left[(p_0 - e_y)^\top \Delta z \right], \quad (5)$$

$$Q(B) = \text{mean} [\text{KL}(p_0 \| p_t)] \geq 0. \quad (6)$$

The equality follows from the log-partition identity and is used as an arithmetic check. It is not a forecast because it uses the observed final logits.

To connect the loss change to parameter motion, recompute the batch gradient at θ_0 and define

$$A(B) = \nabla_{\theta} \ell_{\theta_0}(B)^\top (\theta_t - \theta_0), \quad R(B) = W(B) - A(B). \quad (7)$$

The exact change can then be written $L_t - D = A + R + Q$. The residual R includes the nonlinearity of the parameter-to-logit map along the observed displacement, as well as any replay or numerical error; we do not identify it with a single unmeasured Hessian term.

The second-order logit approximation to the convex log-partition term is

$$Q_2(B) = \frac{1}{2} \text{mean}_{(x,y) \in B} [\text{Var}_{v \sim p_0(\cdot|x)} (\Delta z_v(x))]. \quad (8)$$

Our primary local closure is therefore $L_t - D \approx A + Q_2$. Both A and Q_2 use the displacement that actually occurred: A uses $\theta_t - \theta_0$, and Q_2 uses Δz . The test explains a realized finite response; it does not predict a future response from a frozen gradient alone.

We retain the measured remainder as the fourth primary term:

$$\boxed{L_t(B) = D(B) + A_t(B) + Q_{2,t}(B) + \varepsilon_t(B)}, \quad \varepsilon_t = R_t + (Q_t - Q_{2,t}). \quad (9)$$

Adding the observed remainder gives an identity; the substantive approximation is that the remainder is small on the response scale $L - D$. Its magnitude and its two components are tested rather than assumed.

Two sources of local curvature. Let $J_0 = D_{\theta} z(\theta_0)$, $r = p_0 - e_y$, and $C_0 = \text{diag}(p_0) - p_0 p_0^\top$. The exact checkpoint Hessian is

$$H_\ell = J_0^\top C_0 J_0 + \sum_v r_v H_{z_v}. \quad (10)$$

For a small parameter displacement δ , the leading variance term is $Q_2 = \frac{1}{2} \delta^\top J_0^\top C_0 J_0 \delta + O(\|\delta\|^3)$, while $R = \frac{1}{2} \delta^\top (\sum_v r_v H_{z_v}) \delta + O(\|\delta\|^3)$. The full loss curvature therefore contains both probability curvature and network curvature. In particular, ε is generally second order in parameter motion, even though $Q - Q_2$ is third order in the local logit displacement under smoothness and bounded derivatives. We do not equate Q_2 with the full parameter-Hessian sharpness.

An output-head-only continuation gives an exact control. Frozen hidden states make logits affine in the changed output matrix, so $W = A$, $R = 0$, and $\varepsilon = Q - Q_2$ up to numerical precision. This separates network nonlinearity from the log-partition approximation without changing the initial predictive function.

A paired criterion for strong RGI. Let P subtract the mean over a fixed set of prefix–target pairs. Two branches from the same checkpoint satisfy

$$\Delta\ell = \Delta A + \Delta Q_2 + \Delta\varepsilon, \quad P\Delta\ell = P(\Delta A + \Delta Q_2 + \Delta\varepsilon). \quad (11)$$

The shared D cancels. Thus a small raw-loss discrepancy can be dominated by frozen difficulty while a nonzero constant offset still fails: that stronger property requires the centered response in Eq. (11) to be small relative to its mean. This is a measurable criterion, rather than a consequence of the coefficient one in Eq. (9).

For a fixed teacher distribution $q(\cdot \mid x)$, the same algebra replaces e_y by q . With $p_t = \text{softmax}(z_q + v)$, the teacher-forced hard-target loss decomposes as

$$\ell = S + C + E, \quad S = -\log q_y, \quad C = (q - e_y)^\top v, \quad E = \text{KL}(q \parallel p_t). \quad (12)$$

Under a target drawn from q , $\mathbb{E}_q[C \mid x] = 0$, and the population cross-entropy is $H(q) + E$. We use this controlled population objective to separate label sampling from the natural-text case, where the unknown conditional distribution is not observed.

4. Common late-checkpoint protocol

Common model and targets. All six settings load the official Pythia-70M (Biderman et al., 2023) step-143,000 checkpoint, revision `de3e4e2d` . . . , with identical 70,426,624 parameters and initial forward computation. The checkpoint is late Pile pretraining; the continuation distribution is FineWeb (Penedo et al., 2024). We do not assume equilibrium on the latter. We reuse a registered 4,672-document FineWeb sample-10BT pool, split 4,096/64/512 into training, calibration, and evaluation. Each document contributes one 256-token span. Teacher-forced scoring uses 223 next-token targets at positions 33–255 and the full 50,304-token Pythia vocabulary. Selection used no model scores, but the pool was used by previous diagnostics; it is not a fresh held-out confirmation.

Paired restarts. Each setting runs 512 batch-eight updates with zero initial optimizer state and a 32-update linear warmup followed by a constant rate. Three registered permutations use seeds 202610081–202610083. They are paired data-order interventions, not independent datasets. Arriving-batch measurements precede its update. At the shared checkpoint, $A = Q_2 = \varepsilon = 0$. A separate 32-document held-out probe is scored at steps 0, 16, 32, 64, 128, and every 64 steps through 512; its frozen D is constant in time.

The six settings are full-model SGD and Heavy Ball at two effective rates, reset Adam (Kingma and Ba, 2017), and output-head-only SGD at the smaller rate. Heavy Ball uses $\beta = 0.9$, no dampening, no Nesterov update, and raw SGD rate $(1 - \beta)h$, giving the same dc-normalized rate h as SGD. No setting uses weight decay or dropout. Head-only training preserves the initial predictive function and freezes all parameters except the output matrix.

Adam uses betas (0.9, 0.95), $\epsilon = 10^{-8}$, and a scalar rate calibrated before training on the first 16 calibration documents. A reset one-step update is scaled to match the smaller-rate SGD full-vocabulary Q_2 , with at most two square-root ratio corrections. This matches an initial functional step scale, not dc gain, direction, or training progress. Evaluation scores do not enter calibration.

Rate protocols and adaptive chronology. The first registered rates were $h = 3 \times 10^{-4}$ and 1.2×10^{-3} . After their smaller-rate replica-zero SGD/momentum closure failed, a second protocol fixed 3×10^{-6} and 3×10^{-5} . Their replica-zero closures also failed, so a terminal protocol fixed 3×10^{-8} and 3×10^{-7} before its measurements. Every protocol runs all six settings and all three permutations; all results are retained. This is an explicitly adaptive rate study. Smaller and larger identify registered rates within each protocol, not measured stability regimes. Fixed 512-update comparisons do not establish a fixed-effective-progress asymptotic limit.

Table 1: Core four-term table for common step-143,000 starts and 512 arrivals. SGD/HB and head-only use effective rate 3×10^{-8} ; Adam’s calibrated rate is 3.02668×10^{-9} . HB denotes normalized Heavy Ball. Scaled nats are explicit in each row. The statistics deliberately differ: batch standard deviation for D , signed means for A, Q_2 , and RMS for ε . Six-setting tables, including the tenfold larger rate, are retained in the supplement.

Term and statistic	SGD	HB	Adam	Head only
D : batch std. (nats)	0.18530 to 0.18634	0.18530 to 0.18634	0.18530 to 0.18634	0.18530 to 0.18634
A : mean (10^{-4} nats)	−1.78 to −1.76	−1.76 to −1.75	−6.60 to −6.56	−0.95 to −0.94
Q_2 : mean (10^{-5} nats)	1.08 to 1.26	1.07 to 1.26	1.12 to 1.13	0.14 to 0.15
ε : RMS (10^{-5} nats)	2.42 to 3.49	2.43 to 3.53	0.26 to 0.29	$(2.76 \text{ to } 2.86) \times 10^{-4}$

Measurements, precision, and gates. Parameters, gradients, logits, and reductions use FP64; TF32 is disabled. We recompute the frozen batch gradient for A , and calculate W, Q, Q_2 over the complete vocabulary. Batch-level means contain all four primary terms; document records contain L, D, W, Q, Q_2 . At step 512 a full-model JVP supplies token-level A , checked against the independent parameter-gradient dot product. These observed finite displacements explain realized responses rather than forecast unknown updates.

The frozen prediction fraction is $1 - \sum_t (L_t - D_t)^2 / \sum_t (L_t - \bar{L})^2$, with a 90% gate. Response closure uses $\text{RMS}(\varepsilon) / \text{RMS}(L - D)$, with a 10% gate; paired closure uses $\text{RMS}(\Delta\varepsilon) / \text{RMS}(\Delta L)$. Token constant-gap testing uses $\text{std}(\Delta\ell) / |\text{mean}(\Delta\ell)|$ at 10%, with near-zero means flagged separately. Near 0:128, far 384:512, full 0:512, and nonoverlapping 64-step blocks are fixed in advance. Signed covariance shares can be negative because terms cancel and are descriptive rather than independent causal effects. The independent checker reconstructs saved full-vocabulary fixtures, identities, schedules, and hashes; it does not independently replay every network gradient or training step.

5. The four measured curves and their validity range

Table 1 is the central result. It separates the shared batch scale from each training setting’s signed linear work, positive variance cost, and measured remainder. The terminal local-scale protocol is shown first; the earlier failed rate protocols remain in Table 2 and the supplementary four-term figures. Each range spans three paired permutations of one reused data pool, not a confidence interval or independent dataset replications.

Shared difficulty dominates raw loss at the local scale. For the terminal protocol, every setting’s frozen prediction fraction exceeds 99.9926%. The shared batch standard deviation of D is 0.18530–0.18634 nats, while mean work and variance are orders of magnitude smaller. This establishes dominance of frozen batch difficulty in these raw arriving losses. It does not establish stochastic independence, a natural Bayes entropy, or a common optimizer-response coefficient.

Response closure is conditional, even at small rates. Adam passes the full-window response gate at 0.3387–0.3748% relative RMS. The affine head-only control passes at 0.002410–0.002470%. SGD and normalized HB narrowly fail at 10.51–14.96% and 10.56–15.21%, respectively. Their near windows pass (SGD 1.89–7.13%, HB 1.82–7.12%), but far windows fail (SGD 12.14–18.44%, HB 12.15–18.77%). Thus small update size is not sufficient for a uniformly small remainder across an accumulated 512-update displacement. These failures are retained rather than treating the raw loss’s near-perfect agreement as a passed mechanism test.

In all terminal settings, A has a negative mean and Q_2 a positive mean. On the full raw response, the signed covariance shares for low SGD are 1.007–1.033 for A , −0.0457 to −0.0347 for Q_2 , and 0.0131–0.0322 for ε . For Adam they are 1.020–1.021, −0.0235 to −0.0230, and 0.00263–0.00279. Linear work dominates response variation and curvature offsets some of it. These descriptive signed shares account for cancellation and do not identify independent causal percentages.

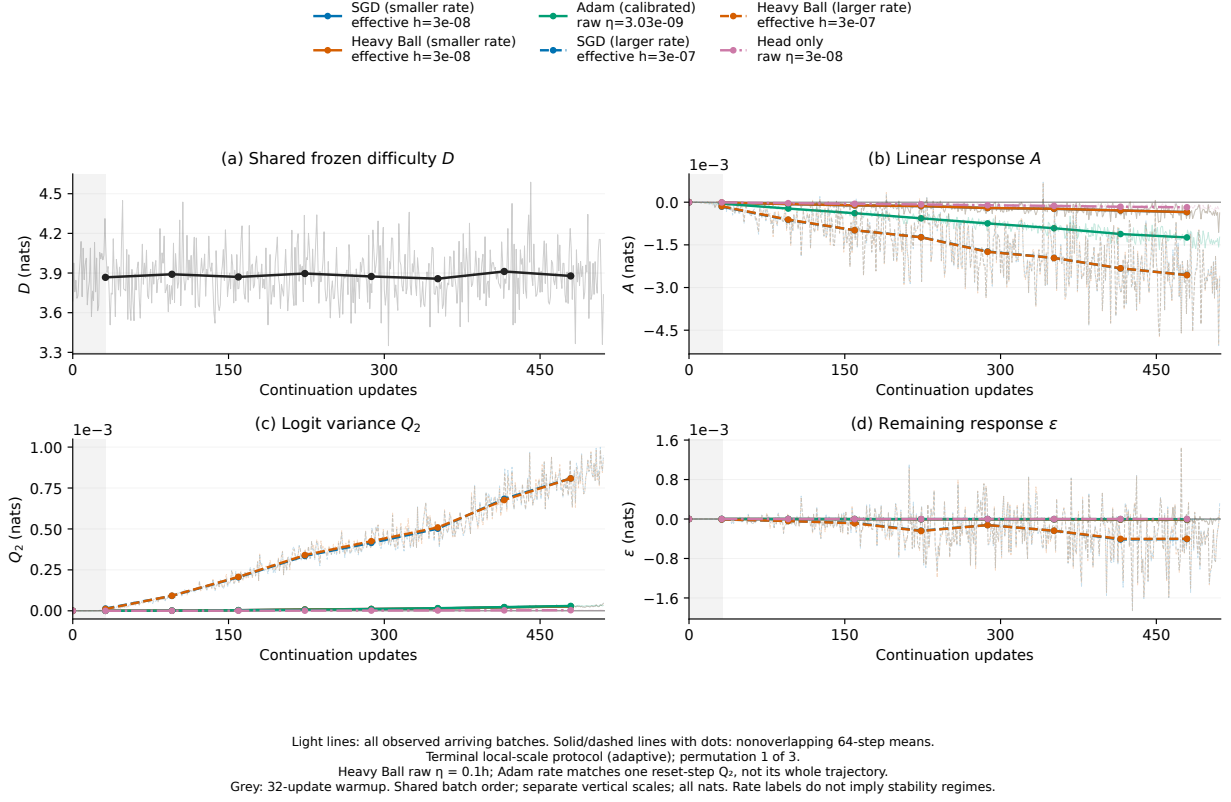


Figure 1: Four terms on the common arriving-batch stream, with permutation zero fixed before results. The shared frozen D is drawn once. Thin raw traces and prespecified 64-step block means are retained; all panels use nats with independent vertical scales. Shading marks the 32-update warmup. A, Q_2, ε begin at zero; D varies because the arriving batch changes. Full six-setting curves are shown, including the tenfold larger rates.

Table 2: Full-window response error $100 \text{ RMS}(\varepsilon) / \text{RMS}(L - D)$, in percent. Ranges span all three permutations. The registered gate is 10%; rates label controlled interventions, not measured stability regimes. The second and third protocols were adaptively registered after failures of the preceding smaller-rate pair.

Setting	Original	Intermediate	Terminal
SGD, smaller rate	129.5–150.7	33.83–44.66	10.51–14.96
HB, smaller rate	200.3–222.7	34.50–37.96	10.56–15.21
Adam, calibrated	178.0–178.7	19.97–20.39	0.3387–0.3748
SGD, larger rate	862.3–1011.3	47.68–48.97	25.54–29.24
HB, larger rate	113.8–127.9	47.82–50.89	26.13–30.04
Head-only SGD	21.49–22.16	2.24–2.42	0.002410–0.002470
Smaller effective rate	3×10^{-4}	3×10^{-6}	3×10^{-8}
Larger effective rate	1.2×10^{-3}	3×10^{-5}	3×10^{-7}

The output-head control identifies the network remainder. In the terminal protocol, full-model SGD’s $\text{RMS}(R)$ is 2.419×10^{-5} – 3.480×10^{-5} nats, versus only 1.349×10^{-7} – 2.588×10^{-7} for $\text{RMS}(Q - Q_2)$. HB has the same ordering. Head-only R is below 1.91×10^{-14} nats in both smaller-rate protocols, as predicted by its affine logit map. In the intermediate protocol, head-only closure passes at 2.24–2.42%, while full SGD/HB fail at 33.83–44.66% and 34.50–37.96%. The contrast separates network nonlinearity from the softmax approximation error using a function-preserving initial intervention.

The original large-rate settings also expose the probability-curvature boundary. High SGD has mean $Q_2 = 11.33$ – 13.10 nats and a softmax-remainder RMS of 15.72–19.27 nats. Calibrated Adam has a softmax-remainder RMS of 0.3140–0.3154 nats. In these cases Q_2 is not an accurate

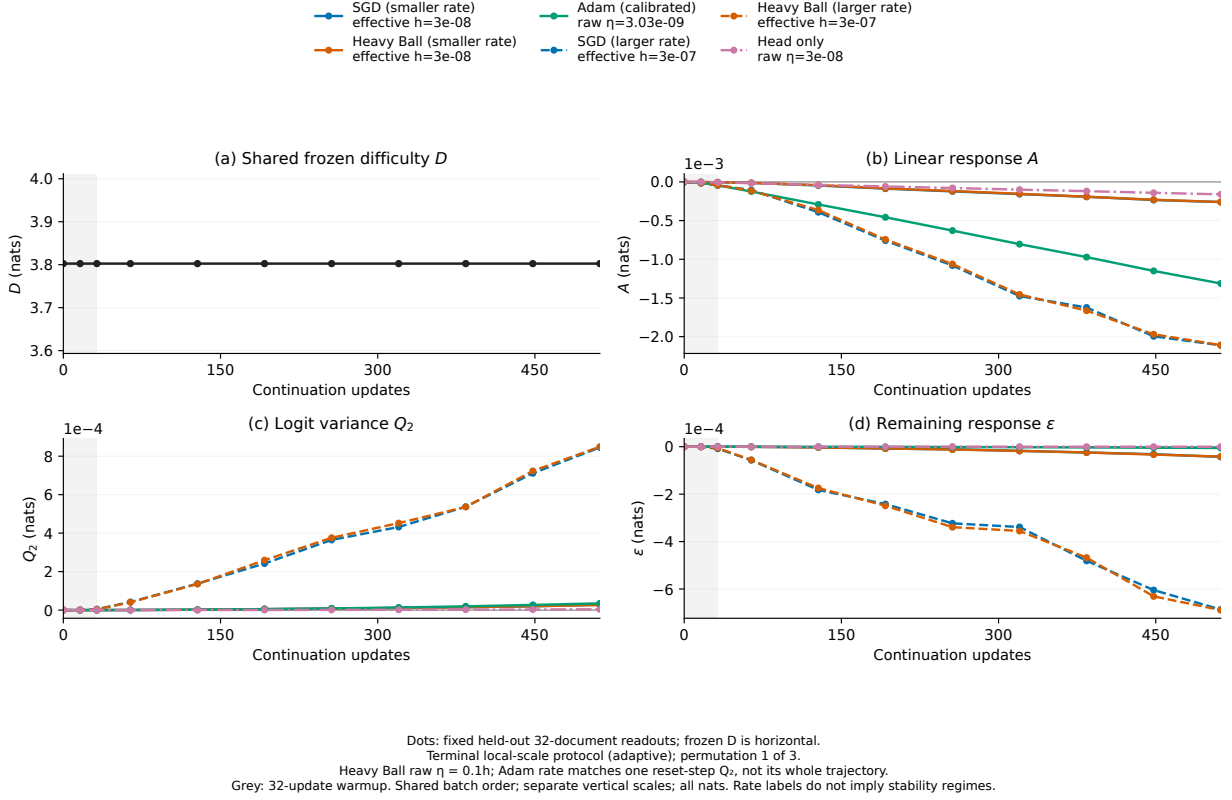


Figure 2: The same decomposition on a fixed 32-document held-out probe. Frozen D is a horizontal line: its arriving-stream fluctuation in Fig. 1 is batch composition, not a time-varying frozen predictor. The other terms are scored at the prespecified readouts. No curve is aligned by a fitted time warp.

substitute for the exact nonnegative Q . This does not violate Eq. (6); it falsifies the useful quadratic approximation in that displacement regime.

Parameter alignment does not establish a constant token gap. At the terminal smaller rate, SGD/HB endpoint displacement cosine is 0.999706–0.999809 and relative distance 2.00–2.43%. At the intermediate smaller rate the cosine is 0.99107–0.99500; at the original smaller rate it is 0.73536–0.75361. This supports better finite-time alignment as the registered rate decreases, at a fixed update count. It is an endpoint displacement comparison, not a claim that every point on the trajectory coincides.

Even at the terminal smaller rate, paired arrival response closure fails at 22.18–28.30%. On the final probe, SGD/HB token gap has centered RMS 1.460×10^{-4} – 2.936×10^{-4} nats and mean magnitude 1.569×10^{-6} – 3.141×10^{-6} . The std/absolute-mean ratio is 46.49–163.48, far above 0.1. The SGD/Adam ratio is 8.362–8.470. Nearly aligned parameter motion and near-identical raw losses therefore coexist with a heterogeneous small paired gap. These continuations do not establish the stronger nonzero constant-offset RGI property.

6. Same-stream momentum alignment and its limits

The common-checkpoint design admits an exact comparison that does not require slowly varying stochastic batch gradients. Let $s_{t+1} = s_t - h_t g_t(s_t)$ be SGD and $m_{t+1} = m_t - h_t v_t$ normalized Heavy Ball on the same prescribed sequence of batch objectives, with $v_t = \beta v_{t-1} + (1 - \beta) g_t(m_t)$, $v_{-1} = 0$, and $s_0 = m_0$. Writing $e_t = m_t - s_t$ and $k = \beta/(1 - \beta)$, summation by parts gives the

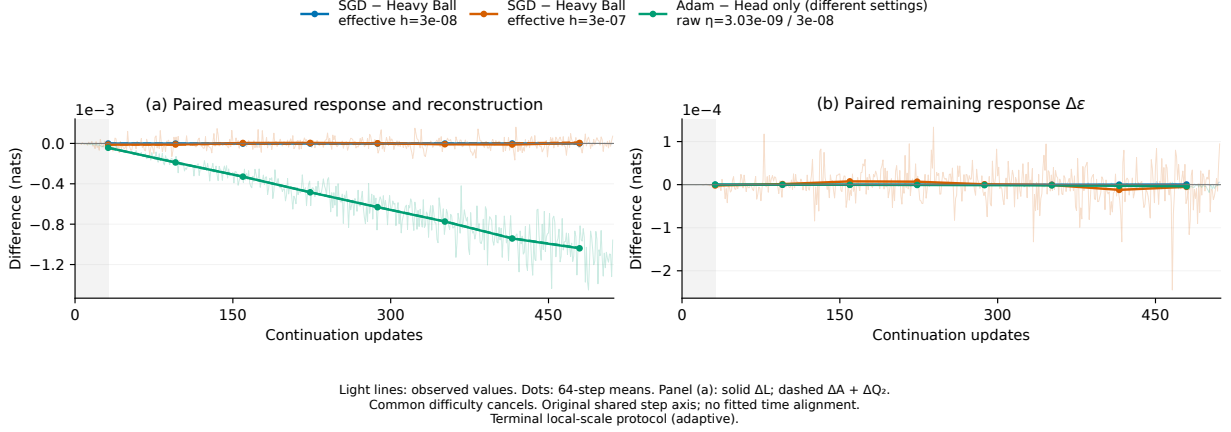


Figure 3: Paired response and measured remaining discrepancy after the exactly shared D cancels. Raw values and prespecified block means are retained. This scale reveals sample-conditioned differences that a raw-loss plot can hide.

exact finite-time identity

$$e_T = - \sum_{t < T} h_t \{g_t(m_t) - g_t(s_t)\} + k \left[h_{T-1} v_{T-1} - \sum_{t < T-1} (h_{t+1} - h_t) v_t \right]. \quad (13)$$

The first term is gradient feedback from the divergent parameters; the second is an explicit memory and warmup contribution. In particular, reset momentum has a genuine startup transient even at matched dc gain. We preserve it in the measured curves.

If every batch gradient is Λ -Lipschitz on a common region containing both paths and $\|g_t(m_t)\| \leq G$, a nondecreasing warmup to $h_{\max}$ gives

$$\|e_T\| \leq 2 \frac{\beta}{1 - \beta} h_{\max} G \exp \left(\Lambda \sum_{t < T} h_t \right). \quad (14)$$

Indeed $\|v_t\| \leq G$, the bracket in Eq. (13) has norm at most $2h_{\max}G$, and discrete Gronwall yields the result. With a constant rate the leading two can be replaced by one. This establishes an alignment limit as $h_{\max} \rightarrow 0$ at fixed integrated progress under uniform regularity. The bound can be loose, and our measurements do not estimate Λ or G ; a same-update-count rate contrast is not a numerical proof of the fixed-progress asymptotic statement.

On a fixed probe, $|\Delta A| \leq \|g_B(\theta_0)\| \|e_T\|$. For the frozen categorical covariance seminorm $\|\cdot\|_{C_B}$,

$$|\Delta Q_2| \leq \frac{1}{2} \|\Delta z_a - \Delta z_b\|_{C_B} (\|\Delta z_a\|_{C_B} + \|\Delta z_b\|_{C_B}). \quad (15)$$

Actual function alignment consequently constrains the first two response differences; $\Delta \varepsilon$ and centered token gaps must still be checked. These equations do not cover Adam or imply a sample-independent nonzero offset.

7. Limits and conclusion

The common-checkpoint experiment supports a concrete local explanation and identifies its limits. Frozen difficulty dominates raw batch fluctuation at sufficiently small registered rates. Negative parameter work is the main source of learning-response variation, while positive logit variance offsets part of it. The full-model remainder contains network curvature that is removed by the affine output-head control; large displacements also invalidate the softmax quadratic approximation. These are separately measured terms, not an attribution inferred from correlated losses.

The strongest desired conclusion remains obstructed. Low-rate SGD/HB endpoint displacements nearly align, but accumulated response closure can fail and their small token-wise gap is not a nonzero constant offset. Even the terminal smaller-rate SGD/HB full-window errors exceed the fixed 10% gate. Adam and the head-only control pass in that setting. We do not shrink the rate further in this revision or relabel failed settings as validation of a universal coefficient. The exact decomposition and the alignment bound organize the evidence; neither makes its approximation gates pass automatically.

The study uses Pythia-70M late-Pile-checkpoint continuations on a reused FineWeb pool, three permutations rather than independent datasets, reset optimizer states, and fixed 512-update rather than matched effective-progress horizons. The second and third rate protocols are explicitly adaptive. Initial Adam curvature matching is not a match of direction or learning speed. All finite-response terms use observed displacements. The independent checker reconstructs recorded arithmetic and saved fixtures; it does not independently replay every training trajectory. The original RGI paper’s larger, from-scratch pretraining setting may consequently occupy a different regime.

The result is a falsifiable mechanism study: it locates a shared batch difficulty, a method-dependent first-order response, two distinct curvature sources, and the measured terms that prevent a stronger constant-gap inference. A future explanation of strong RGI must additionally make the centered paired response small on the relevant token support, and verify that condition on an untouched window or independent data.

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A. Reproducibility details

Data and model hashes. The FineWeb source is `HuggingFaceFW/fineweb`, configuration `sample-10BT`, revision prefix `9bb295dd`, shard `sample/10BT/001_00000.parquet`. The pool selection starts after source row 65,536 and uses selection seed 610000 and split seed 610001. The verified tokenizer commit begins `a39f36b1`. Full hashes and input manifests are in [analysis/rgi_extension_20261006/new-provenance.json](#) and the accompanying evidence manifest in the research workspace.

Evaluation units. Natural batches contain 8 documents and 223 scored next-token targets per document (query positions 32 through 254). Difficulty and response statistics are computed from document-level means; token positions within a document are not treated as independent

documents. The known-teacher continuation uses 33 natural seed tokens followed by 63 sampled tokens and evaluates 63 targets per generated document.

Verification scope. Each of the three common-start protocols completed all 18 runs and passed 2,278 independent checks, for 6,834 total. The respective maximum numerical errors are 3.35×10^{-13} , 9.59×10^{-14} , and 2.39×10^{-14} . NumPy reconstructs the final eight-token full-vocabulary fixtures independently, and checks zero initial terms, paired document difficulty, fixed-probe difficulty, schedules, hashes, response identities, and JVP mean parity. All 54 fixtures were harvested with hash verification. The checker does not independently replay network gradients or training. A separate CPU FP64 real-model pulse agrees between gradient-dot and JVP work to 6.25×10^{-16} , providing numerical engineering evidence rather than an independent full training replication.

The original protocol ran on A100 with PyTorch 2.7.0; the two rate extensions ran on H200 with PyTorch 2.9.1. Comparisons are paired within each protocol. Checkpoint canonical tensor bytes match across hosts; small cross-host operation differences are not assumed absent.

Additional diagnostics. The registered 64-step block analysis uses signed covariance shares; these shares are attribution coefficients and may exceed one or be negative because terms cancel. High-pass SGD failures and the optimizer-difference failure are retained in the main text rather than averaged away. The three permutations reuse one data pool and are not independent datasets.

Code and artifacts. The arXiv source package includes the central table, four-term curves, retained failed rate protocols, MetaCircle assets, and compiled bibliography. The public project provides compact numerical evidence, protocol contracts, independent checker and plotting code. The release manifest identifies which raw weights, corpus spans, and full logit fixtures are omitted; a saved-evidence audit is distinct from rerunning training.

B. Retained four-term curves outside the terminal scale

C. Historical distinct-start diagnostics

The earlier study used optimizer-specific continuation starts and a common step-32,000 scoring reference. It is retained as historical evidence and does not replace the common late-checkpoint experiment. Its main scales were $\text{std}(D) = 0.1843\text{--}0.1937$, mean $A = -0.004011\text{--}0.003552$ (SGD) and $-0.003290\text{--}0.003272$ (Adam), mean $Q_2 = 0.001744\text{--}0.001992$ and $0.000622\text{--}0.000634$, and $\text{RMS}(\varepsilon) = 0.000191\text{--}0.000286$ and $0.0000754\text{--}0.0000838$ nats. These small-remainder values are not transferred to the later common checkpoint or newly calibrated rates.

D. Known-teacher continuation

Historical protocol. This supporting experiment predates the common step-143,000 study. The teacher is official Pythia-70M step 32,000, and each student starts at official step 16,000. From 33 natural seed tokens, the teacher samples 63 tokens at temperature one over the full 50,304-token vocabulary. The document split is 1,024/64/128 for training/calibration/evaluation. Hard sampled targets and full soft teacher distributions are compared over three paired permutations, 128 updates, and batch size eight. Optimizer states are reset; SGD uses rate 10^{-3} and Adam uses 10^{-6} with $\beta = (0.9, 0.95)$ and $\epsilon = 10^{-8}$. These settings and starts are distinct from the new late-checkpoint main experiment.

The natural experiment has no observed conditional distribution, so its reference decomposition cannot decide whether the target residual is label noise or a systematic model mismatch. The known-teacher experiment removes that ambiguity. For the fixed teacher q , the population objective is exactly $H(q) + \text{KL}(q\|p)$, while hard targets additionally expose the sampled terms S and C in Eq. (12).

The teacher entropy averages 4.1798397 nats, the sampled teacher surprisal averages 4.1723279

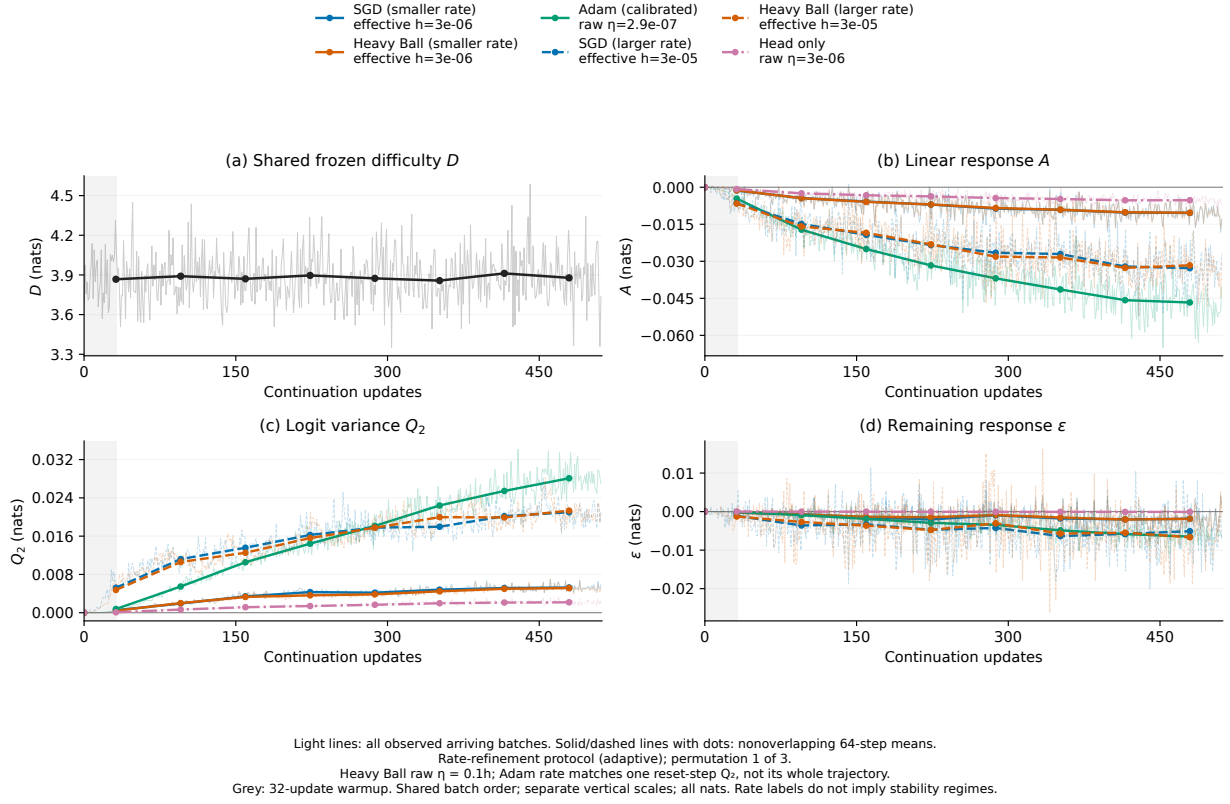


Figure 4: Intermediate adaptive protocol, $h = 3 \times 10^{-6}/3 \times 10^{-5}$. The same shared checkpoint and batch sequence are used. Complete four-term curves retain the full-model network remainder and the passing affine head-only control.

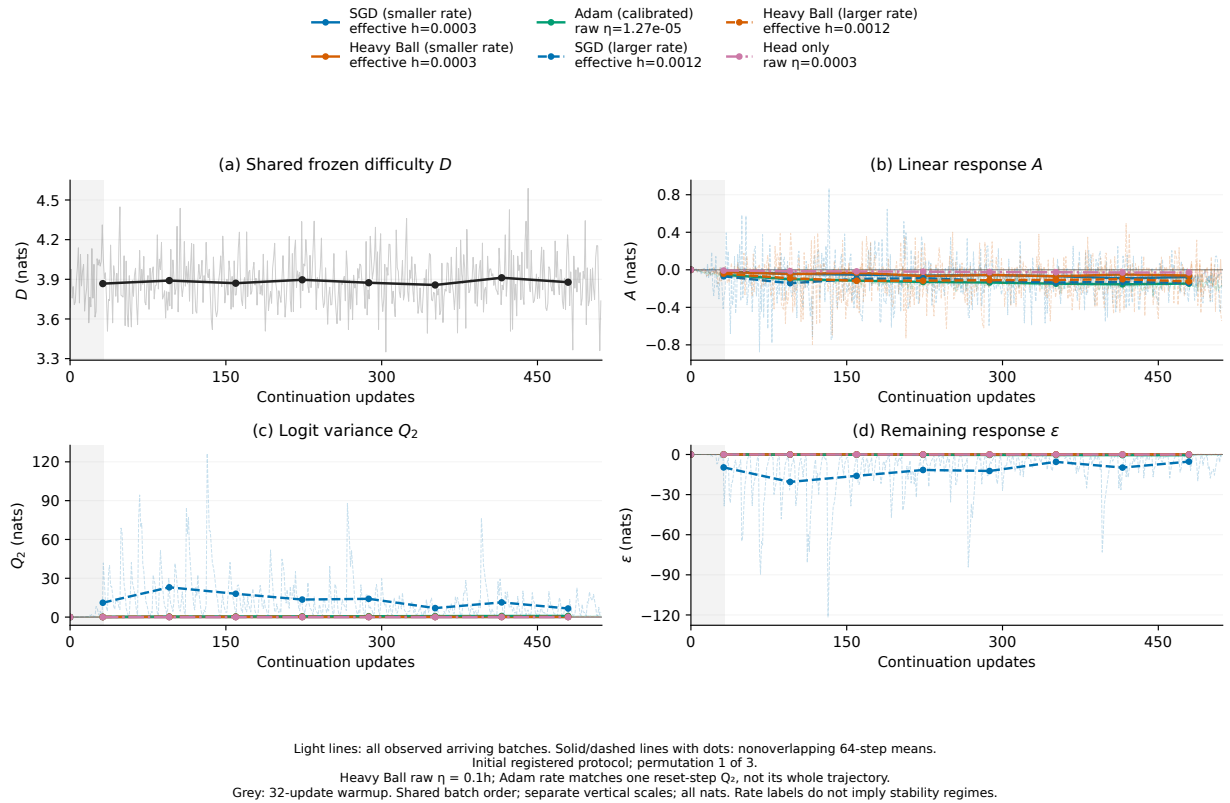


Figure 5: Original protocol, $h = 3 \times 10^{-4}/1.2 \times 10^{-3}$. The high-SGD variance and softmax approximation error are retained, not clipped or removed. The figure exhibits an approximation boundary; it does not establish an edge-of-stability regime.

Table 3: All six terminal settings, complete numerical ranges in nats.

Setting	D : std.	A : mean	Q_2 : mean	ε : RMS
SGD, smaller rate	0.18530 to 0.18634	-1.7783×10^{-4} to -1.7613×10^{-4}	1.0801×10^{-5} to 1.2629×10^{-5}	2.4217×10^{-5} to 3.4880×10^{-5}
HB, smaller rate	0.18530 to 0.18634	-1.7593×10^{-4} to -1.7481×10^{-4}	1.0746×10^{-5} to 1.2552×10^{-5}	2.4265×10^{-5} to 3.5303×10^{-5}
Adam, calibrated	0.18530 to 0.18634	-6.5959×10^{-4} to -6.5632×10^{-4}	1.1175×10^{-5} to 1.1333×10^{-5}	2.6141×10^{-6} to 2.8720×10^{-6}
SGD, larger rate	0.18530 to 0.18634	-0.0014754 to -0.0014446	3.4178×10^{-4} to 3.9758×10^{-4}	3.9843×10^{-4} to 4.6482×10^{-4}
HB, larger rate	0.18530 to 0.18634	-0.001473 to -0.0014394	3.4408×10^{-4} to 4.0682×10^{-4}	4.0713×10^{-4} to 4.7802×10^{-4}
Head-only SGD	0.18530 to 0.18634	-9.4748×10^{-5} to -9.3932×10^{-5}	1.4350×10^{-6} to 1.4546×10^{-6}	2.7625×10^{-9} to 2.8632×10^{-9}

Table 4: All six intermediate adaptive settings, complete ranges in nats.

Setting	D : std.	A : mean	Q_2 : mean	ε : RMS
SGD, smaller rate	0.18530 to 0.18634	-0.0072221 to -0.0070185	0.00329 to 0.003695	0.0018671 to 0.0025149
HB, smaller rate	0.18530 to 0.18634	-0.0071508 to -0.0070151	0.0033529 to 0.0035083	0.001909 to 0.0021383
Adam, calibrated	0.18530 to 0.18634	-0.031325 to -0.031151	0.015665 to 0.015712	0.0041303 to 0.0042013
SGD, larger rate	0.18530 to 0.18634	-0.023052 to -0.022785	0.015396 to 0.015453	0.0064298 to 0.0066531
HB, larger rate	0.18530 to 0.18634	-0.02311 to -0.022634	0.01503 to 0.015307	0.0065332 to 0.0069865
Head-only SGD	0.18530 to 0.18634	-0.003818 to -0.0037352	0.001405 to 0.0014572	6.2245×10^{-5} to 6.8153×10^{-5}

Table 5: All six original settings, complete ranges in nats.

Setting	D : std.	A : mean	Q_2 : mean	ε : RMS
SGD, smaller rate	0.18530 to 0.18634	-0.059062 to -0.053999	0.092781 to 0.094434	0.040653 to 0.044295
HB, smaller rate	0.18530 to 0.18634	-0.065866 to -0.049914	0.084416 to 0.087984	0.04319 to 0.05079
Adam, calibrated	0.18530 to 0.18634	-0.12508 to -0.12293	0.57904 to 0.57999	0.32982 to 0.33144
SGD, larger rate	0.18530 to 0.18634	-0.11471 to -0.1083	11.333 to 13.095	15.715 to 19.275
HB, larger rate	0.18530 to 0.18634	-0.12516 to -0.068801	0.21556 to 0.25371	0.16121 to 0.20456
Head-only SGD	0.18530 to 0.18634	-0.020902 to -0.020852	0.016448 to 0.016703	0.0017283 to 0.001781

nats, and the initial teacher KL of the student probe is 0.2593348 nats. After 128 updates, teacher KL ranges are 0.229442–0.229879 for hard SGD, 0.241563–0.241657 for hard Adam, 0.228137–0.228251 for soft SGD, and 0.229840–0.229902 for soft Adam. The population $A + Q_2$ closure error is at most 2.795% across the tested horizons, objectives, and permutations.

The term means show why a KL-only explanation is incomplete. In hard-teacher SGD at 128 updates, the mean linear work is about -0.057 nats, curvature is about $+0.027$, and the net risk change is about -0.030 . In hard-teacher Adam the corresponding values are about -0.022 , $+0.004$, and -0.018 . Soft targets change both magnitudes. The cancellation is a finite-displacement observation and does not establish that two optimizers share a common first-order coefficient.

The same control also tests strong RGI. After exact teacher-population integration, the cross-prefix standard deviation of the hard optimizer gap divided by its absolute mean is 3.59–3.69; the soft value is 10.69–11.89. Both are far above a strict 0.1 constant-offset threshold. Removing sampled-label noise therefore does not recover a uniform optimizer shift across prefixes.

E. A sufficient support-gate construction

For completeness, consider a prefix x with a known support S_x , a normalized shape p_x on that support, and a reject token. Let

$$q_h(y \mid x) = \alpha(h)p_x(y) \quad (y \in S_x), \quad q_h(\text{reject} \mid x) = 1 - \alpha(h), \quad (16)$$

where $\alpha(h) = \text{sigmoid}(h)$. For every target in the support,

$$\ell_h(x, y) = -\log p_x(y) - \log \alpha(h), \quad \partial_h \ell_h = \alpha(h) - 1. \quad (17)$$

Thus all target losses share the same algorithm-dependent offset, even though the common target surprise varies with $p_x(y)$. A dog/cat numerical check gives a gap 0.5901707965 with standard deviation 1.05×10^{-16} across five prefixes and both targets.

This is a sufficient support-gate mechanism, not evidence about natural LLMs. It relies on a common normalized shape inside a prefix-dependent support and on a shared scalar mass gate. When the evaluated target support covers the full vocabulary, normalization prohibits an exact nonzero uniform loss shift over every target: no outside probability mass remains for this construction. Measuring whether natural models have an approximate candidate support and a common mass gate is a separate experiment.